

Chapter 24 — Basic Principles of Immunology and Immunotherapy in Urologic Oncology

- **Immunotherapy is an established modality** in bladder, kidney, lung cancer and melanoma; its **long history in bladder cancer** is underappreciated.
- **Bacille Calmette-Guérin (BCG)** — a relatively non-specific agent whose mechanism is still incompletely understood — is the **model framework for genitourinary (GU) immunotherapy**.
- **Cancer vaccines** aim instead to induce a **specific** anticancer response (prostate, kidney).
- **Immune checkpoint blockade (ICB)**: antigen-specific responses are restrained by checkpoint molecules on CD4 and CD8 **tumour-infiltrating lymphocytes (TILs)**, typified by **programmed death-1 (PD-1)** and **programmed death ligand-1 (PD-L1)**. Blocking antibodies give **objective responses (tumour shrinkage)** in **about 20%** of kidney or bladder cancers.
- ICB is **more effective in combination**; **anti-cytotoxic T-lymphocyte antigen-4 (CTLA-4)**, which probably targets suppressive CD4 cells, is a key combination partner.

INNATE IMMUNE SYSTEM

- **Innate vs adaptive**: innate is evolutionarily older (all vertebrates), rapid, pattern-based; adaptive is specific and has memory.
- Innate recognition: **pathogen-associated molecular patterns (PAMPs)** bind **Toll-like receptors (TLRs)** (homologues of *Drosophila* Toll) — the fundamental mechanism by which the organism senses "danger".
- **BCG cell-wall peptidoglycans are canonical PAMPs → bind TLR2** on innate cells in the bladder wall → multistep immune response.
- **First responder: the tissue-resident macrophage** (*makros* large, *phagos* eat) — engulfs pathogens; on PAMP recognition secretes cytokines and chemokines that recruit and activate further cells.
- **Neutrophil (polymorphonuclear neutrophil, PMN)**: **most abundant circulating immune cell (about 60% of blood leucocytes)**; half-life hours in blood but days in inflamed tissue; **main constituent of pus, hallmark of acute inflammation**. Emigrates between endothelial cells up a chemokine gradient — the **hypersegmented nucleus eases deformation**. Releases secretory granules on PAMP recognition.

TABLE 24.1 SELECTED CELL TYPES INVOLVED IN THE IMMUNE RESPONSE TO GENITOURINARY CANCERS

Cell	Arm	Role
Epithelial / urothelial cell	Sentinel	Secretes type I interferons and chemokines in response to stress, inflammation, viral infection or PAMP danger signals
Macrophage	Innate	Engulfs pathogens and dead cells; cytokines/chemokines initiate or amplify the response
Neutrophil (PMN)	Innate	Most numerous innate cell in blood; controls bacterial infection; pus = acute inflammation
Dendritic cell (DC)	Bridge	Presents peptides from dead cells/debris to T cells in the lymph node; activated by PAMPs or cytokines
CD4 T cell	Adaptive	"Helper" — helps CD8 cells kill or B cells secrete antibody
CD8 T cell	Adaptive	"Killer" — once activated, serially lyses specific targets
Regulatory T cell (Treg)	Adaptive	CD4 subset defined by transcription factor FoxP3 ; downregulates responses — detrimental in cancer

- **Pattern**: sentinel → innate effector → bridge (DC) → adaptive effector → brake (Treg). **Trap**: the Treg is a CD4 cell, marked by FoxP3, and its suppressive function works against the patient in cancer.

CYTOKINES AND CHEMOKINES

- **Cytokine:** any small, immunologically relevant secreted messenger (*cyto* cell, *kinesis* movement) — how epithelium signals to immune cells and immune cells to one another.
- **Type I interferons (IFN- α , IFN- β):** from virally infected or stressed epithelium → make cells **more visible to adaptive immunity** (upregulate class I major histocompatibility complex [MHC] and antigen processing) and directly promote epithelial death. **Intravesical IFN- α : encouraging but mixed results** in bladder cancer.
- **Chemokine:** cytokine subset whose main role is **migration along a concentration gradient**. Prototype **CXCL8 (interleukin-8, IL-8)** — secreted by urothelium; **powerful neutrophil attractant**; probably important in the BCG response.
- **Interleukins** are numbered by order of discovery, so the name says little about function. **IL-1 is really an innate cytokine:** from stressed epithelium, attracts immune cells, and systemically is a **principal mediator of fever**.
- **Th1 vs Th2 polarisation** of CD4 T cells decides the direction of the adaptive response (Table 24.2). In an **IL-12-rich milieu** naive CD4 cells become **Th1**; with **IL-4** they become **Th2** (chronic inflammation, antibody).
- **Th1 → successful BCG response; Th2 → BCG failure.** Cytokines are **detectable systemically after BCG**; **raised serum IL-2 after treatment predicts a favourable outcome** — BCG effects are not purely local.

TABLE 24.2 SELECTED CYTOKINES, CHEMOKINES, AND INTERLEUKINS INVOLVED IN THE IMMUNE RESPONSE TO GENITOURINARY CANCERS

Group	Molecule	Source	Role
Type I interferon	IFN- α , IFN- β	Urothelium, epithelium, macrophages	Secreted by virally infected or TLR-activated cells; upregulate class I MHC and antigen processing
Chemokine	CXCL8 (IL-8)	Urothelium, epithelium	Powerful neutrophil chemoattractant
Chemokine-like	IL-1	Epithelium, macrophages	Attracts other cells, e.g. monocytes from blood
Chemokine	SLC (secondary lymphoid-tissue chemokine, CCL21)	Lymph-node stromal cells	Draws activated DCs and T cells into nodes; sensed by CCR7
Polarising	IL-12	DCs	Naive CD4 → Th1
Polarising	IL-4	DCs, natural killer T cells	Naive CD4 → Th2
Th1 output	IL-2, TNF- α , IFN- γ	Th1 cells	Favourable BCG response ; drive CD8 killing
Th2 output	IL-4, IL-5, IL-10, IL-13	Th2 cells	Unfavourable BCG response ; drive antibody production

- The table tests **which signal polarises (IL-12 vs IL-4)** and **which outputs mark success (Th1) or failure (Th2)**. **Trap:** IL-4 is both the Th2-inducing signal and a Th2 product; IL-2 belongs to Th1. TNF = tumour necrosis factor.

Concept — *Th1 is cellular and cytotoxic, hence antitumour; Th2 is humoral and chronic-inflammatory, hence BCG failure. The cytokine milieu at the moment of CD4 priming decides which.*

ADAPTIVE IMMUNE SYSTEM

- **CD4 helper, CD8 killer and B cells**, recruited after innate activation; they confer **specificity and memory** (stronger response on re-encounter).

- **Dendritic cell — the bridge from innate to adaptive.** Named for fine processes resembling neurones; resident in tissues (e.g. **Langerhans cells** in skin); at rest samples its surroundings by **pinocytosis** and stays quiescent without a danger signal.
- **DC activation sequence:** danger signal (TNF- α from macrophages, or bacterial products via TLRs) → **stops antigen uptake, retracts dendrites** → **upregulates MHC** (binds 9–12-amino-acid peptides) and **costimulatory B7-1 and B7-2** → **migrates via afferent lymphatics following secondary lymphoid chemokine through CCR7** → activates specific CD4, CD8 and B cells in the node.
- **Why migration is needed:** resting lymphocytes live in lymph nodes; resting DCs live in tissues.
- **CD4 activation:** DC presents cognate peptide on **class II MHC**. **Th1 fully activates CD8 killers; Th2 helps B cells mature into plasma cells.**
- **Tregs** suppress adaptive responses and impede a Th1/CD8 anticancer response. **Natural** Tregs arise in the thymus; **induced** Tregs form when naive CD4 cells meet antigen in a low-inflammation, **transforming growth factor- β (TGF- β)**-rich milieu. Laboratory data favour **natural, thymus-derived Tregs** as the critical population.
- **CD8 T cell:** recognises **about 9-amino-acid peptides on class I MHC**, present on almost all cells and upregulated by inflammation and viral infection. Killing is **exquisitely specific** (type 1 diabetes: β cells lysed, adjacent α cells spared), works by **inducing apoptosis** in the target, and is **serial**. CD8 cells are activated by vaccines and **augmented by checkpoint blockade**.

MHC CLASS I VS CLASS II

	Class I	Class II
Recognised by	CD8 (killer)	CD4 (helper)
Distribution	Almost all cells; $\uparrow$ by inflammation, viral infection, type I IFN	Antigen-presenting cells (presented by DCs)
Peptide length	About 9 amino acids	Longer (typically 13–25 amino acids)

IMMUNE EDITING HYPOTHESIS

- Apart from virally driven tumours of the immunocompromised, **human cancers arise in immunologically intact hosts** — so host and tumour interact throughout progression.
- **Three phases** (the three Es): **Elimination** (innate macrophages and DCs with adaptive cells destroy small, undetectable tumours) → **Equilibrium** (tumour acquires genetic, epigenetic and stromal changes; growth slowed but not eliminated; may last the host's lifetime) → **Escape** (clinically apparent tumour).
- **Escape mechanisms:** downregulation of targeted tumour antigens; **downregulation of MHC**; induction or expansion of **Tregs**.
- **Subversion of antitumour immunity is now a formal hallmark of cancer.**

Concept — A clinically evident cancer is by definition an **escaped cancer**; immunotherapy works by reversing the escape mechanism that that tumour happens to depend on.

KEY POINTS: BASIC IMMUNOLOGY

- Innate response first (rapid, non-specific) → adaptive (specific, memory).
- Antitumour response needs **Th1: IFN- γ , IL-2, TNF- α** .
- CD4+ Tregs inhibit the adaptive response.
- Immune editing: early tumours eliminated; clinical tumours must escape.

CHRONIC INFLAMMATION AND THE ENDOGENOUS IMMUNE RESPONSE TO GENITOURINARY CANCERS

- **Apparent paradox:** immune editing says antitumour immunity helps, yet **chronic inflammation promotes tumours** — clearly in bladder cancer, probably in prostate cancer.

Chronic Inflammation and the Immune Response to Bladder Cancer

- **Strongest GU evidence linking inflammation to carcinogenesis.**
- *Schistosoma haematobium* is an International Agency for Research on Cancer (IARC) known carcinogen: endemic countries have high bladder cancer rates; heavy infestation → **squamous cell phenotype**.
- Other inflammatory causes: **chronic urinary tract infection, chronic indwelling catheters, cyclophosphamide cystitis.**
- Presumed mechanism (as in other cancers): dysfunctional **M2 macrophages** secreting immunosuppressive cytokines; **myeloid-derived suppressor cells (MDSCs)**; **Th2 and Treg polarisation.**
- Supporting immune editing: **CD8 infiltration correlates with better outcome in muscle-invasive bladder cancer (MIBC).**
- **Escape via checkpoints:** checkpoint molecules on tumour-specific T cells bind ligands on tumour cells or tumour-associated macrophages → proliferation and effector function strongly inhibited.
- **Bladder tumours express PD-L1;** expression rises with grade, was **more prognostic than World Health Organization (WHO) grade**, tracks immune infiltration, and is **high in BCG-induced granulomas of patients progressing on BCG** — a possible escape route.
- **Adaptive immune resistance:** tumour mutations → CD8 recognition → CD8 cells infiltrate and secrete **IFN-γ** → IFN-γ **induces PD-L1** on tumour, myeloid and epithelial cells → PD-L1 engages **PD-1** on the same CD8 cells → effector function curtailed. Hence anti-PD-1/PD-L1 antibodies should work — confirmed in bladder and kidney cancer.
- **Trap:** PD-L1 is largely **induced by the immune attack itself** — a marker of an ongoing but suppressed response, not a fixed tumour trait.

IMMUNE CHECKPOINT MOLECULES AT THE T CELL–TUMOUR/APC INTERFACE (FIG. 24.4)

T-cell molecule	Ligand on tumour cell or antigen-presenting cell (APC)	Effect
T-cell receptor (TCR)	MHC–peptide	Signal 1 — specificity
CD28	CD80 / CD86 (B7)	Signal 2 — costimulation (activating)
CTLA-4	CD80 / CD86	Inhibitory; blocked by anti-CTLA-4 antibody
PD-1	B7-H1 (= PD-L1)	Inhibitory; blocked by anti-PD-1 antibody
LAG-3	MHC class II	Inhibitory

Immune Microenvironment in Kidney Cancer

- **Inflammation–kidney cancer link is weak**, suggested only by proinflammatory risk factors (**smoking, obesity**).
- **The renal paradox:** in most cancers ↑ CD8 density → better prognosis; in **renal cell carcinoma (RCC)** ↑ **CD8 density** → **worse outcome**, in primary and metastatic lesions; mechanism unknown.
- **Worst subgroup:** high CD8 infiltration + high PD-1 and **lymphocyte activation gene-3 (LAG-3)** + few functional antigen-presenting cells.

- Deep flow-cytometric profiling of primary tumours: marked complexity (**17 macrophage and 22 T-cell subsets**); **PD-1 broadly expressed**, LAG-3 and TIM-3 narrowly; a **CD38-high macrophage population** travels with PD-1+ T cells → **CD38 blockade is a combination candidate** (anti-CD38 antibodies are approved or in development).

Chronic Inflammation and the Immune Response to Prostate Cancer

- Macrophages, T and B cells infiltrate the prostate **even without symptoms or cancer**; this baseline inflammation is probably tumour-promoting.
- **Three lines of evidence**: (1) inflammation concentrates in the **peripheral zone, where more than 90% of tumours arise**; (2) **Western populations (endemic prostatic inflammation) have higher incidence than Asian populations**; (3) **proliferative inflammatory atrophy (PIA)** — flattened but proliferating epithelium with inflammatory cells — lies **adjacent to high-grade prostatic intraepithelial neoplasia (PIN)**, the most convincing link.
- Adaptive environment **dominated by Tregs** — in the gland, in neoplastic lesions, and as functional Tregs in **bone marrow**.
- **CD8 cells can be suppressive (pro-tumour)**; regulatory CD8 cells in blood can mask an antigen-specific response.
- Infiltrating CD8 cells are **PD-1-positive and clonally restricted**.

ENDOGENOUS IMMUNE LANDSCAPE: BLADDER VS KIDNEY VS PROSTATE

	Bladder	Kidney	Prostate
Inflammation → cancer link	Strong (schistosomiasis, UTI, catheters, cyclophosphamide)	Weak (smoking, obesity only)	Probable (peripheral zone, geography, PIA)
Meaning of CD8 infiltration	Better outcome (MIBC)	Worse outcome (paradox)	CD8 cells may be suppressive
Dominant restraint	PD-L1 on tumour, ↑ with grade	PD-1/LAG-3-high T cells, few APCs; CD38 macrophages	Tregs ; PD-1+ CD8 cells

Concept — *Chronic, Th2/Treg/M2-skewed inflammation promotes cancer, whereas an acute Th1/CD8 response eliminates it. The three GU tumours differ in which arm dominates, and that predicts where checkpoint blockade later succeeds (bladder, kidney) and fails (prostate).*

KEY POINTS: CHRONIC INFLAMMATION AND THE ENDOGENOUS IMMUNE RESPONSE TO GU CANCERS

- Bladder cancer is promoted by chronic inflammation from infection or other stimuli.
- Kidney: an active CD8 response exists, but higher CD8 density → worse prognosis.
- Prostate cancer may also be driven by chronic inflammation.
- Checkpoint molecules on TILs attenuate the adaptive response to GU tumours.

IMMUNOTHERAPY FOR GENITOURINARY CANCERS

Bacille Calmette-Guérin in Bladder Cancer

- **BCG sequence**: instil $1-5 \times 10^8$ **viable mycobacteria** → most washed out at first void; a fraction **adheres to urothelium via fibronectin attachment protein** → cell-wall PAMPs stimulate urothelium through **TLR2 and TLR4** (direct macrophage/DC activation also likely) → urothelium secretes **IL-8** (neutrophil attractant) and **TNF-α** (recruits and activates macrophages) → **within hours, a neutrophil wave** enters the bladder wall, clears bacilli and releases cytokines that activate DCs → **over weeks, an adaptive, mainly CD4-driven**

response: DCs prime CD4 cells in draining nodes → **Th1 (IL-2, TNF- α , IFN- γ)** → consolidation as **long-lived memory**.

- Memory is unproven in patients but implied by the **~50% of non-muscle-invasive disease remaining in remission for years** after BCG.

Concept — *Successful BCG therapy is a **typical immune response**: innate (neutrophils, macrophages) first, then adaptive Th1 CD4, then memory.*

Nadofaragene Firadenovec in Bladder Cancer

- **Background:** intravesical recombinant **IFN- α 2b protein** is well tolerated and active in non-muscle-invasive bladder cancer (NMIBC) after BCG, inducing apoptosis via **TNF-related apoptosis-inducing ligand (TRAIL)** pathways — but **responses are not durable**, largely because dwell time is only 1–2 hours.
- **Gene therapy rationale:** maximise transgene expression in urothelium, minimise adenovirus outside the bladder. Early adenoviral attempts failed against the bladder's **antiadherence barrier**; the excipient **Syn3 markedly enhances transduction**.
- **Product:** non-replicating recombinant adenovirus carrying the human IFN- α 2b gene + Syn3.
- **Landmark phase III, BCG-unresponsive NMIBC:** 75 mL intravesical (3×10^{11} viral particles/mL) every 3 months for up to 12 months; 1-hour dwell with patient rotating; cytology and cystoscopy ($\pm$ biopsy) every 3 months. Primary endpoint: complete response (CR) in carcinoma in situ (CIS) $\pm$ high-grade Ta/T1.
- **Results:** CR 53.4% within 3 months (CIS cohort); median CR duration 9.69 months; 24.3% high-grade recurrence-free at 12 months. No grade 4–5 drug-related events; commonest: discharge around catheter, fatigue, bladder spasm, urgency.
- **US Food and Drug Administration (FDA) approval 16 December 2022** — first gene therapy for BCG-unresponsive NMIBC with CIS $\pm$ papillary tumours.

Cancer Vaccines

- Aim, as with BCG: raise a T-cell response. **Two components:** an **adjuvant** (PAMP-like; activates DCs — "adds" immunogenicity) and a **target antigen** overexpressed in tumour relative to normal tissue (choice largely empirical).
- **Sequence:** intradermal injection → immature DC takes up antigen; adjuvant activates it → DC migrates to node, presents peptide on MHC → TCR recognition (signal 1) **plus costimulation (signal 2) from a mature DC** is required → CD8 cells gain lytic function against cells bearing the same MHC-peptide; CD4 cells become Th1 → CD8 cells leave the node and traffic widely.

Vaccines for Kidney Cancer and Vaccines for Prostate Cancer

PHASE III CANCER VACCINE TRIALS IN GU CANCER

Vaccine	Setting	Platform	Comparator	Verdict
IMA901	Metastatic RCC, human leucocyte antigen (HLA)-A*02+	Tumour-associated peptides + GM-CSF adjuvant; low-dose cyclophosphamide to deplete Tregs	Sunitinib $\pm$ vaccine	No overall survival (OS) benefit ; non-significant trend favouring no vaccine
AGS-003	Metastatic high-risk RCC	Autologous DCs (leukapheresis; monocytes matured with IL-4 + GM-CSF) transfected with tumour RNA ; intranodal	Sunitinib $\pm$ vaccine	Terminated for lack of efficacy

Vaccine	Setting	Platform	Comparator	Verdict
Sipuleucel-T	Asymptomatic / minimally symptomatic metastatic castration-resistant prostate cancer (mCRPC)	Autologous peripheral blood mononuclear cells incubated ~36 h with PA2024 (prostatic acid phosphatase [PAP]-GM-CSF fusion) ; IV every 2 weeks × 3	Placebo (2:1)	OS +4.1 months (25.8 vs 21.7); no effect on time to progression; first FDA-approved vaccine for a solid tumour
PROSTVAC-VF	Asymptomatic / minimally symptomatic mCRPC	Heterologous poxviral prime–boost (vaccinia prime, fowlpox boost) encoding prostate-specific antigen (PSA) + TRICOM (B7.1, LFA-3, ICAM-1)	Placebo vs vaccine ± GM-CSF	Closed for futility (OS)

- **Granulocyte-macrophage colony-stimulating factor (GM-CSF)** — a strong inducer of DC migration — is the recurring adjuvant.
- **IMA901 failure fits the renal paradox**: more CD8 infiltration may not help in RCC.
- **Autologous vaccines** are limited by variable tumour material; **RNA extraction** lets a small resected mass yield ample vaccine. Patients' own DCs are often dysfunctional, hence **ex vivo DCs generated outside the tolerogenic environment**.
- Sipuleucel-T product is **heterogeneous** (mature APCs plus T, B and NK cells); it primes PAP-specific CD4+ and CD8+ cells as a **prime–boost** — the first infusion primes, later ones boost.
- **Pattern**: only sipuleucel-T succeeded; every other GU phase III vaccine failed. **Trap**: sipuleucel-T prolongs survival **without** delaying progression.

PD-1/PD-L1 Blockade in Genitourinary Cancers

- The **most important evasion mechanism**: T-cell PD-1 binding PD-L1 limits proliferation and killing.

ANTI-PD-1/PD-L1 AND ANTI-CTLA-4 ANTIBODIES: TARGETS

Agent	Target	Antibody
Nivolumab	PD-1	Fully human IgG4
Pembrolizumab	PD-1 (blocks binding of PD-L1 and PD-L2)	Humanised, hinge-stabilised IgG4
Avelumab	PD-L1	IgG1
Atezolizumab, durvalumab	PD-L1	—
Ipilimumab	CTLA-4	—

PD-1/PD-L1 Blockade in Bladder Cancer

- Bladder cancer frequently expresses PD-L1; anti-PD-1/PD-L1 agents are approved in **non-muscle-invasive, muscle-invasive and advanced/metastatic urothelial cancer (mUC)**.
- **Baseline for advanced disease**: most progress within **~9 months** of platinum chemotherapy; median OS **14–15 months** with cisplatin regimens — the rationale for maintenance.

ANTI-PD-1/PD-L1 TRIALS IN UROTHELIAL CANCER

Setting / trial	Agent and population	Result	Status
NMIBC KEYNOTE-057	Pembrolizumab 200 mg IV 3-weekly up to 24 months; BCG-unresponsive CIS ± papillary , ineligible for or declining cystectomy	CR 41% at 3 months ; median CR duration 16.2 months; 19% in CR ≥12 months; grade 3–4 events 13%	FDA January 2020

Setting / trial	Agent and population	Result	Status
Adjuvant MIBC CheckMate 274	Nivolumab 2-weekly up to 1 year vs placebo; high risk after radical surgery: pT3, pT4a or pN+ (no neoadjuvant cisplatin and ineligible for or declining adjuvant cisplatin) or ypT2-ypT4a or ypN+ (after neoadjuvant cisplatin)	Disease-free survival (DFS) 20.8 vs 10.8 months ; better non-urothelial-tract recurrence-free and distant metastasis-free survival; grade ≥ 3 events 17.9% vs 7.2%	FDA August 2021
Adjuvant IMvigor010	Atezolizumab vs observation	DFS not improved	Negative
Adjuvant AMBASSADOR	Pembrolizumab vs observation	Ongoing	—
First line, cisplatin-ineligible IMvigor210 → IMvigor130	Atezolizumab (phase II) → atezolizumab + chemotherapy (confirmatory)	Accelerated approval April 2017; IMvigor130 missed OS co-primary	Withdrawn
First line, cisplatin-ineligible KEYNOTE-052	Pembrolizumab, phase II, any PD-L1 status	Objective response rate (ORR) 24% overall, 28% PD-L1+; accelerated approval May 2017. June 2018: combined positive score (CPS) <10 → worse OS than carboplatin chemotherapy	Restricted: cisplatin-ineligible with CPS ≥ 10 , or ineligible for any platinum regardless of PD-L1
Maintenance JAVELIN Bladder 100	Avelumab 2-weekly + best supportive care vs best supportive care; no progression on first-line platinum	OS longer overall and in PD-L1+; grade ≥ 3 events 47.2% vs 25.2%	FDA June 2020
Second line CheckMate 275	Nivolumab 3 mg/kg 2-weekly, single-arm phase II, post-platinum	ORR 20.7%; median response duration 20.3 months; tumour mutational burden and PD-L1 predicted response	Accelerated February 2017
Second line KEYNOTE-045	Pembrolizumab 200 mg 3-weekly vs investigator's chemotherapy (paclitaxel, docetaxel or vinflunine), post-platinum	OS 10.3 vs 7.4 months (P = .002) ; ORR 21.1% vs 11.4% (P = .001)	Only anti-PD-1/PD-L1 agent approved on randomised phase III data
Second line JAVELIN Solid Tumor	Avelumab, phase I expansion; progression on/after platinum or within 12 months of (neo)adjuvant platinum	ORR 16.5%; response duration 20.5 months	Accelerated May 2017
Second line	Atezolizumab, durvalumab	Failed confirmatory trials	Withdrawn

- **Why CheckMate 274 succeeded and IMvigor010 failed is unclear: trial design, or target — PD-1 (nivolumab) vs PD-L1 (atezolizumab).**
- **Pattern:** every surviving bladder indication uses anti-PD-1 except avelumab (maintenance, second line); both other anti-PD-L1 agents were withdrawn. **Trap:** single-agent first-line pembrolizumab is **not** for all cisplatin-ineligible patients — only CPS ≥ 10 or platinum-ineligible.

PD-1/PD-L1 Blockade in Kidney Cancer

- RCC is traditionally **immune-sensitive**, so single-agent anti-PD-1 responses in phase I were expected.
- **CheckMate 025** (first approval): nivolumab 3 mg/kg vs the **mammalian target of rapamycin (mTOR) inhibitor everolimus** in metastatic RCC (mRCC) progressing on first-line tyrosine kinase inhibitor (TKI); primary endpoint OS → **longer OS and fewer grade 3–4 events**; updated median OS **25.8 vs 19.7 months**.
- **KEYNOTE-564** (adjuvant): 1 year pembrolizumab vs placebo in **clear-cell RCC at high risk — pT2 with nuclear grade 4 or sarcomatoid differentiation, $\geq$ pT3, N+, or M1 with no evidence of disease**.

Recurrence or death 32% lower (hazard ratio [HR] 0.68); 24-month DFS 77.3% vs 68.1%. FDA November 2021.

PD-1/PD-L1 Blockade in Prostate Cancer

- Phase II pembrolizumab in docetaxel- and endocrine-pretreated mCRPC (PD-L1+, PD-L1- and bone-predominant cohorts): **poor — PSA response <10%, ORR <5%.**
- **Why:** prostate cancer cells **generally do not express PD-L1**, although most infiltrating CD8 cells express PD-1.
- **Microsatellite instability-high (MSI-H) / mismatch repair-deficient (dMMR)** tumours carry many neoantigens → more immunosensitive. **FDA accelerated approval, 23 May 2017: pembrolizumab for unresectable or metastatic MSI-H/dMMR solid tumours** (adult and paediatric) progressing after prior treatment with no satisfactory alternative — **the first approval based on a biomarker rather than site of origin.**
- **National Comprehensive Cancer Network (NCCN)** prostate guideline: consider MSI-H/dMMR testing; pembrolizumab for MSI-H/dMMR mCRPC **second line or beyond.**
- MSI-H/dMMR prevalence in prostate cancer **1.2–12%**; in one series about one-fifth of these carried a **germline Lynch syndrome** mutation, and **nearly half derived durable benefit** — in line with other MSI-H tumours.

Combination Immunotherapy Regimens in Genitourinary Cancers

CTLA-4 Blockade in Prostate and Kidney Cancer

- CTLA-4 is **relatively overexpressed on FoxP3+ CD4 Tregs**, unlike PD-1 → blockade probably **attenuates Treg function**, freeing CD8 TILs.
- **Ipilimumab monotherapy in prostate cancer:** PSA response ~15–20%, few radiographic responses; **two phase III trials (pre- and post-chemotherapy mCRPC) — no OS benefit.**
- **Rationale for combination:** PD-1 on CD8 killers + CTLA-4 on CD4 Tregs → synergy; the **first checkpoint combination synergistic in animals.**
- **CheckMate 214:** nivolumab + ipilimumab vs sunitinib, first-line clear-cell mRCC; co-primary endpoints in intermediate/poor risk: median OS not reached vs 26.0 months (HR 0.63, $P < .001$); ORR 42% vs 27%; progression-free survival (PFS) 11.6 vs 8.4 months ($P = .03$ — not significant against the prespecified .009 threshold) → **FDA-approved first line for intermediate/poor-risk mRCC.**

PD-1 VS CTLA-4 BLOCKADE

	PD-1 axis	CTLA-4
Main expressing cell	CD8 TILs	FoxP3+ CD4 Tregs
Ligand	PD-L1 (B7-H1), PD-L2	CD80/CD86
Effect of blockade	Releases CD8 effector brake in the tumour	Attenuates Treg suppression
Prostate monotherapy	Poor, except MSI-H/dMMR	PSA responses only; no OS benefit

Combining PD-1/PD-L1 Blockade With Androgen-Deprivation Therapy in Prostate Cancer

- **Androgen-deprivation therapy (ADT) is proinflammatory:** the thymus involutes with age; **androgen ablation restores thymic size and cellularity** → **more naive T-cell output** (aged mice, presumed in humans); ADT before prostatectomy → **activated CD4+ infiltration** of the gland; **new antibody specificities** appear.
- **Enzalutamide raises PD-L1** → phase II adding pembrolizumab on enzalutamide progression: **about one in five had PSA decline ≥50%**; phase III ongoing.

- **IMbassador250 (phase III): atezolizumab + enzalutamide — no OS benefit** in unselected mCRPC; longer PFS with high PD-L1 (IC2/3), CD8 expression and immune gene signatures → **patient selection needed**.

Combining PD-1/PD-L1 Blockade With Chemotherapy in Bladder Cancer

- Chemotherapy is not always immune-antagonistic: some agents enhance vaccine responses by inducing **immunogenic cell death** — tumour cells die in a way that lets resting DCs take up antigen, traffic to nodes and prime tumour-specific T cells.
- **Antibody–drug conjugates (ADCs)**: cytotoxic payload linked to a tumour-specific antibody → less systemic toxicity, higher tumour drug concentration.
- **Enfortumab vedotin (EV)**: anti-nectin-4 antibody (highly expressed in urothelial cancer) + **monomethyl auristatin E** (microtubule disruptor); **first ADC FDA-approved for locally advanced/mUC, December 2019**.
- **EV + pembrolizumab: accelerated approval April 2023 for cisplatin-ineligible locally advanced/mUC** (EV-103/KEYNOTE-869): **ORR 68%, CR 12%** — the **first anti-PD-1 + ADC approval** in the USA.
- Phase III trials adding PD-1/PD-L1 blockade to first-line **gemcitabine–cisplatin** are ongoing; success would make first-line therapy chemo-immunotherapy but leave the best second line uncertain.

Combining PD-1/PD-L1 Blockade With VEGF Inhibition in Kidney Cancer

- **Vascular endothelial growth factor (VEGF)** signalling drives RCC; VEGF-pathway TKIs have long been a mainstay.
- **VEGF is also immunosuppressive**: it impairs **DC maturation** and **T-cell entry** into tumour (altered endothelial trafficking). VEGF inhibition → **vascular normalisation** → **more intratumoural CD8+ cells** and better DC function → rationale for adding ICB.
- **Four PD-1/PD-L1 + TKI combinations are FDA-approved first line for mRCC with a clear-cell component; all beat sunitinib**, with striking ORR and PFS gains.

TABLE 24.3 FDA-APPROVED PD-1/PD-L1 BLOCKADE AND VEGF INHIBITOR COMBINATION THERAPY IN METASTATIC RENAL CELL CARCINOMA

Trial	Combination	ORR: combination vs sunitinib	Median PFS, months: combination vs sunitinib
KEYNOTE-426	Pembrolizumab + axitinib	60% vs 40%	15.4 vs 11.1
JAVELIN Renal 101	Avelumab + axitinib	52.5% vs 27.3%	13.3 vs 8.0
CLEAR	Pembrolizumab + lenvatinib (third arm lenvatinib + everolimus: 53.5%; 14.7)	71% vs 36.1%	23.9 vs 9.2
CheckMate 9ER	Nivolumab + cabozantinib	55.7% vs 27.1%	16.6 vs 8.3

- **Sunitinib is the control in all four**; every combination wins on ORR and PFS. **CLEAR (pembrolizumab + lenvatinib) has the highest ORR and longest PFS**; **avelumab + axitinib is the only anti-PD-L1 pairing**. The table reports ORR and PFS, not OS.

KEY POINTS: PD-1/PD-L1 BLOCKADE AND COMBINATION REGIMENS IN GU CANCERS

- Pembrolizumab: BCG-unresponsive CIS (cystectomy-ineligible/declined); first-line mUC if CPS ≥ 10 and cisplatin-ineligible, or platinum-ineligible; second-line mUC; adjuvant high-risk clear-cell RCC; MSI-H/dMMR mCRPC second line or beyond.
- Nivolumab: adjuvant MIUC after radical surgery; second-line mUC; mRCC after TKI.
- Avelumab: maintenance after first-line platinum; second-line mUC.
- Nivolumab + ipilimumab: first-line intermediate/poor-risk mRCC.

- EV + pembrolizumab: cisplatin-ineligible locally advanced/mUC.
- Four TKI + PD-1/PD-L1 combinations: first-line clear-cell mRCC. ADT + ICB (prostate) and chemo + ICB (bladder) under trial.

Concept — *Every successful GU immunotherapy either **supplies a danger signal** that starts a Th1 response (BCG, vaccine adjuvant, immunogenic cell death, VEGF inhibition) or **removes a brake** on a response already under way (PD-1, CTLA-4). Failures cluster where the brake is not the limiting step — PD-L1-negative, Treg-dominated prostate cancer.*

CONCLUSIONS

- Progress has been especially strong in GU cancers, driven by a better grasp of basic immunology and tumour immune resistance.
- **BCG was among the first immune-based treatments in routine practice in any cancer.**
- The future is **combinations**: PD-1 + CTLA-4 and PD-1/PD-L1 + VEGF inhibitors are approved first line in RCC; ADT + ICB (prostate) and chemotherapy + ICB (bladder) are under study.
- **Most patients will still not benefit** → resistance mechanisms must be defined in preclinical models and patients.